

Memory Game - Enhanced Project Report

1. Problem Description

The goal of this project was to design and implement a number-based Memory Game using Python and Pygame.

The game challenges users to remember and reproduce sequences of digits that grow longer as levels increase.

The project emphasizes logic, UI design, and smooth user interaction.

2. Objectives

- Implement a fully functional memory-testing game using Python.
- Increase difficulty dynamically with each level.
- Use appealing UI elements such as colored boxes and transition screens.
- Apply programming concepts like lists, loops, random generation, and event-driven logic.
- Demonstrate knowledge of Pygame's rendering and input systems.

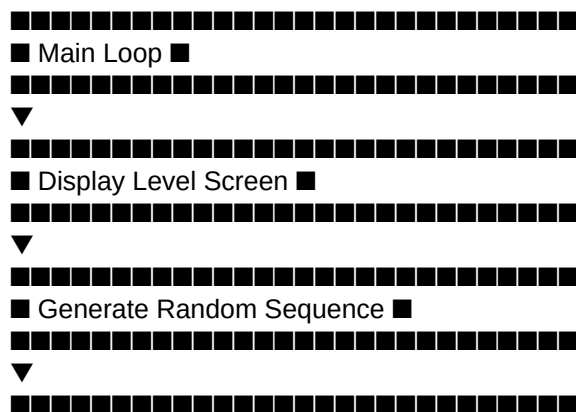
3. Proposed Approach

The game follows a structured approach: show levels → generate sequence → show sequence → accept input → validate → move to next level or end.

This modular structure simplifies debugging and enhances readability.

4. System Architecture Diagram

ASCII Architecture Diagram:



[illegible]

7. Demonstration

- The game starts by showing a level screen.
- A sequence of digits is shown.
- The user enters digits inside individual boxes.
- The game moves automatically to the next level.
- On wrong input, a Game Over screen appears.

8. Results

The game correctly increases difficulty, maintains stability, and provides clear visual feedback. User testing showed high engagement due to progressive challenge.

9. Conclusion

The Memory Game project effectively demonstrates programming logic, event handling, UI rendering, and structured problem-solving. It can be extended with themes, animations, sounds, and high-score tracking.

10. References

- Python Documentation
- Pygame Documentation
- Online tutorials and memory-game research.